

$$\text{cov}(X_1, X_2) = \frac{\sum_{i=1}^n (X_{1i} - \bar{X}_1)(X_{2i} - \bar{X}_2)}{(n-1)}$$

$$\bar{X}_1 = \frac{\sum_{i=1}^n X_{1i}}{n}$$

$$\bar{X}_2 = \frac{\sum_{i=1}^n X_{2i}}{n}$$

$(X_{11}, X_{21}) (X_{12}, X_{22}) \dots (X_{1n}, X_{2n})$ sample
 observations of X_1 & X_2 respectively
 (n of each)

$p(X_1, X_2)$ is the joint distribution

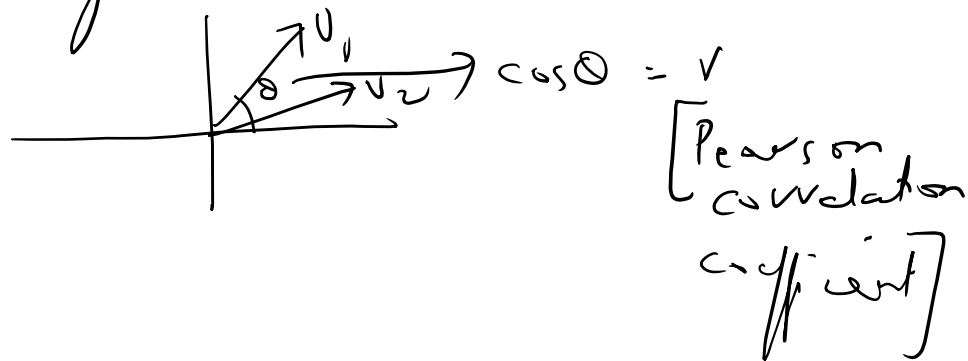
We are sampling this joint distribution
 and drawing n samples

The covariance formula can be interpreted as the dot product between two vectors

$$V_1 = \frac{1}{\sqrt{n-1}} \begin{bmatrix} X_{11} - \bar{X}_1 \\ X_{12} - \bar{X}_1 \\ \vdots \\ X_{1n} - \bar{X}_1 \end{bmatrix} \quad V_2 = \frac{1}{\sqrt{n-1}} \begin{bmatrix} X_{21} - \bar{X}_2 \\ X_{22} - \bar{X}_2 \\ X_{23} - \bar{X}_2 \\ \vdots \\ X_{2n} - \bar{X}_2 \end{bmatrix}$$

n observations of variable X_1

The correlation coefficient is $\cos \theta$ where θ is the angle between these vectors



Correlation coefficient

$$= \frac{S_{X_1, X_2}}{S_{X_1} S_{X_2}} = \frac{U_1 \cdot U_2}{\|U_1\| \|U_2\|}$$

$$S_{X_1, X_2} = \frac{1}{n-1} \sum_{i=1}^n (X_{1i} - \bar{X}_1)(X_{2i} - \bar{X}_2)$$

$$S_{X_1} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_{1i} - \bar{X}_1)^2} \rightarrow \text{length of } U_1$$

$$S_{X_2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_{2i} - \bar{X}_2)^2} \rightarrow \text{length of } U_2$$

$$\text{cov}(X_1, X_1) = \frac{\sum_{i=1}^n (X_{1i} - \bar{X}_1)(X_{1i} - \bar{X}_1)}{n-1}$$

$$= \frac{\sum_{i=1}^n (X_{1i} - \bar{X}_1)^2}{n-1}$$

= sample variance of X_1

$$s_{ij} = \text{Cov}(X_i, X_j)$$

$$s_{ji} = \text{Cov}(X_j, X_i)$$

$$\frac{\sum_{i=1}^n (X_{1i} - \bar{X}_1)(X_{2i} - \bar{X}_2)}{(n-1)}$$

This formula
is symmetric
in X_1 & X_2

$$\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$$

$$E[(X - \mu_x)(Y - \mu_y)]$$

joint probability
density function
of X, Y

$$= \iint (X - \mu_x)(Y - \mu_y) P(X, Y) dx dy$$

double integral

Are the two formulas related?

Sample covariance and the one above?

$$\frac{1}{N-1} \sum_{i=1}^N (x_{1i} - \bar{x}) (x_{2i} - \bar{x}_2)$$

$$\frac{P(x_1, x_2) N}{N}$$

is the number of samples that fall within the joint square

	x_{21}	...	x_{2N}
x_{11}			
x_{12}			
$\vdots$			
x_{1N}			

$$\frac{1}{N-1} \sum_{i=1}^N \sum_{j=1}^N (x_{1i} - \mu_{x_1}) (x_{2j} - \mu_{x_2}) P(x_{1i}, x_{2j}) N$$

In a general time-series
 $\underbrace{\text{Cov}(X_t, X_{t+h})}$ will not necessarily
 be the same as $\underbrace{\text{Cov}(X_2, X_{12})}$

In a stationary time-series

$$\underbrace{\gamma(t, t+h)} = \gamma(h)$$

$$\text{Cov}(X_t, X_{t+h}) = \gamma(h) \text{ independent of } t$$

for observed data $Y_1, Y_2, \dots, Y_T$

calculate the sample mean

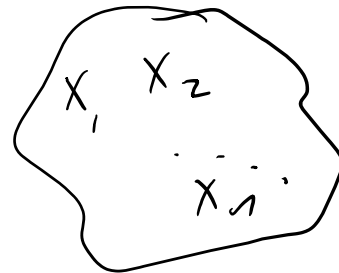
$$\bar{Y} = \frac{1}{T} \sum_{t=1}^T Y_t$$

Sample autocovariance at lag k is estimated as

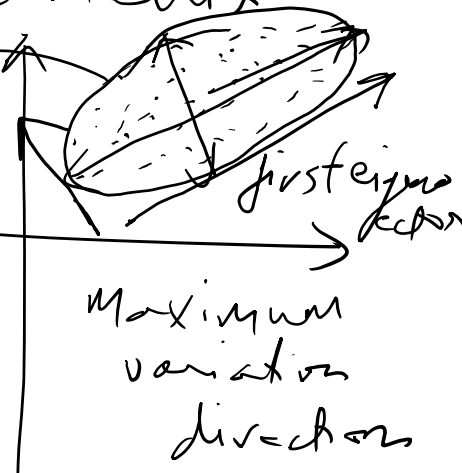
$$\hat{\gamma}(k) = \frac{1}{T} \sum_{t=k+1}^T (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})$$

$$k=5 \Rightarrow \left[\begin{aligned} & (Y_6 - \bar{Y})(Y_1 - \bar{Y}) + (Y_7 - \bar{Y})(Y_2 - \bar{Y}) \\ & + (Y_8 - \bar{Y})(Y_3 - \bar{Y}) + \dots \\ & (Y_T - \bar{Y})(Y_{T-5} - \bar{Y}) \end{aligned} \right]$$

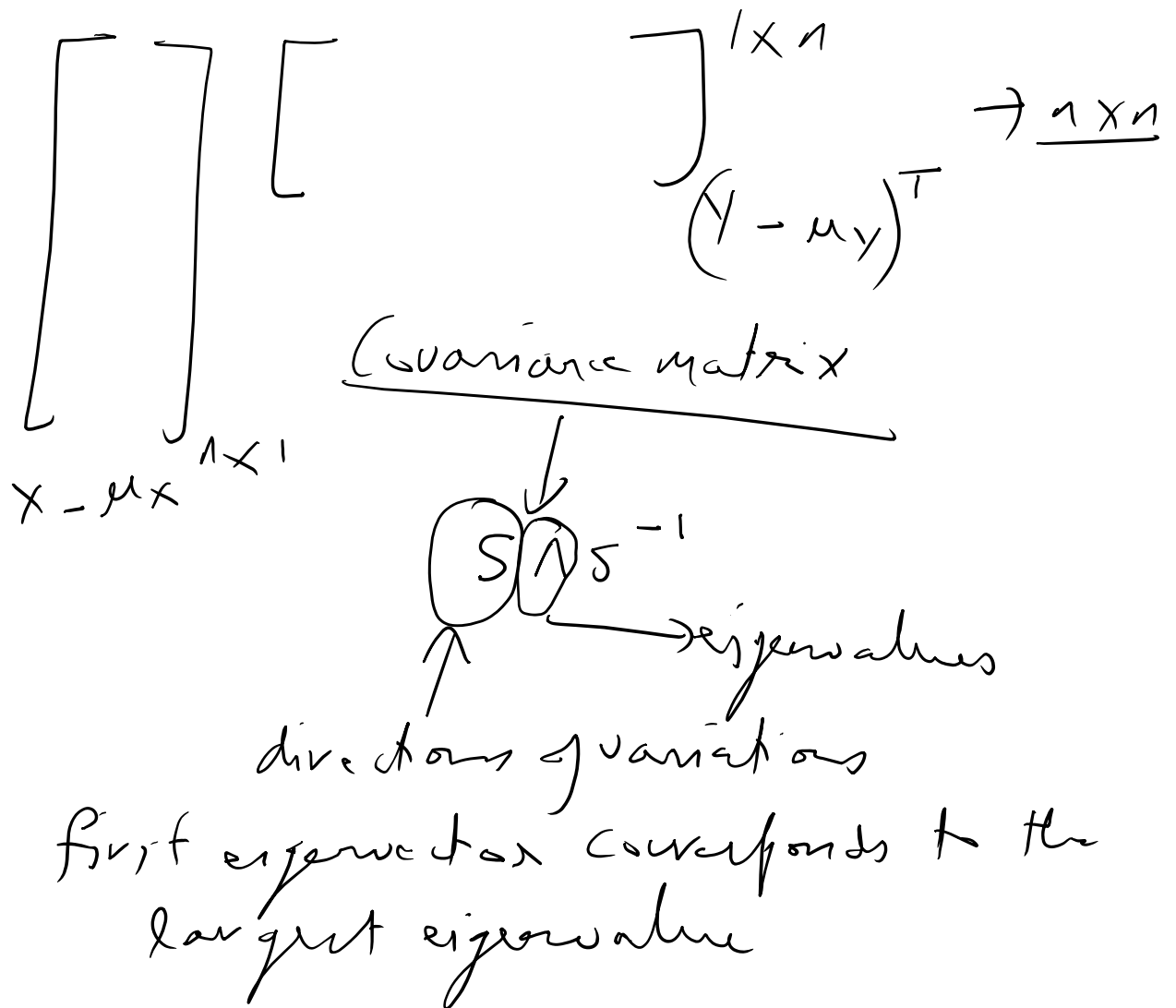
Diagonalizing a covariance matrix



ellipsoid



diagonalize the covariance matrix $E[(X - \mu_X)(Y - \mu_Y)^T]$



$$v_1 = \frac{1}{\sqrt{n-1}} \begin{bmatrix} x_{11} - \bar{x}_1 \\ x_{12} - \bar{x}_1 \\ \vdots \\ x_{1n} - \bar{x}_1 \end{bmatrix}$$

$$v_2 = \frac{1}{\sqrt{n-1}} \begin{bmatrix} x_{21} - \bar{x}_2 \\ x_{22} - \bar{x}_2 \\ \vdots \\ x_{2n} - \bar{x}_2 \end{bmatrix}$$

Let X be one variable and $Y = 10X$

Calculate the correlation coefficient

between X and $Y \rightarrow \underline{+1}$

v_1 and v_2 vectors point in the same direction

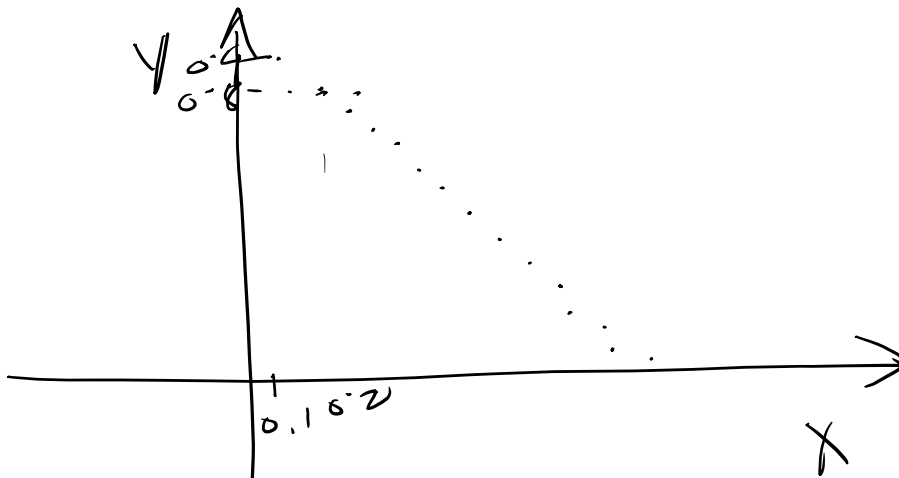
$$r = \frac{v_1 \cdot v_2}{\|v_1\| \|v_2\|}$$

if X_1 and X_2 are independent $v_1 \cdot v_2 \equiv 0$

$$r = \cos \theta = 0 \quad \text{so } \theta = 90^\circ$$

if variables are independent, they are uncorrelated

if variables are uncorrelated, they are not necessarily independent



$$\begin{aligned}(x_1, y_1) &= (0.1, 0.9) \\(x_2, y_2) &= (0.2, 0.8) \\(x_3, y_3) &= (0.3, 0.7)\end{aligned}$$

$$\underline{x + y = 1}$$

$$X - \bar{X} = \begin{bmatrix} 0.1 - 0.5 \\ 0.2 - 0.5 \\ 0.3 - 0.5 \\ \vdots \\ 0.9 - 0.5 \end{bmatrix} = \begin{bmatrix} -0.4 \\ -0.3 \\ -0.2 \\ -0.1 \\ 0 \\ 0.1 \\ 0.2 \\ 0.4 \end{bmatrix}$$

$$\bar{X} = \frac{0.1 + 0.2 + \dots + 0.9}{9} = 0.5$$

$$Y - \bar{Y} = \begin{bmatrix} 0.9 - 0.5 \\ 0.8 - 0.5 \\ \vdots \\ 0.1 - 0.5 \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.3 \\ 0.2 \\ 0.1 \\ 0 \\ -0.1 \\ -0.2 \\ -0.3 \\ -0.4 \end{bmatrix}$$

Now $(X - \bar{X}) = -(Y - \bar{Y})$ they point in opposite directions

the angle between them is 180°

$$r = \cos 180^\circ = -1$$

Regression

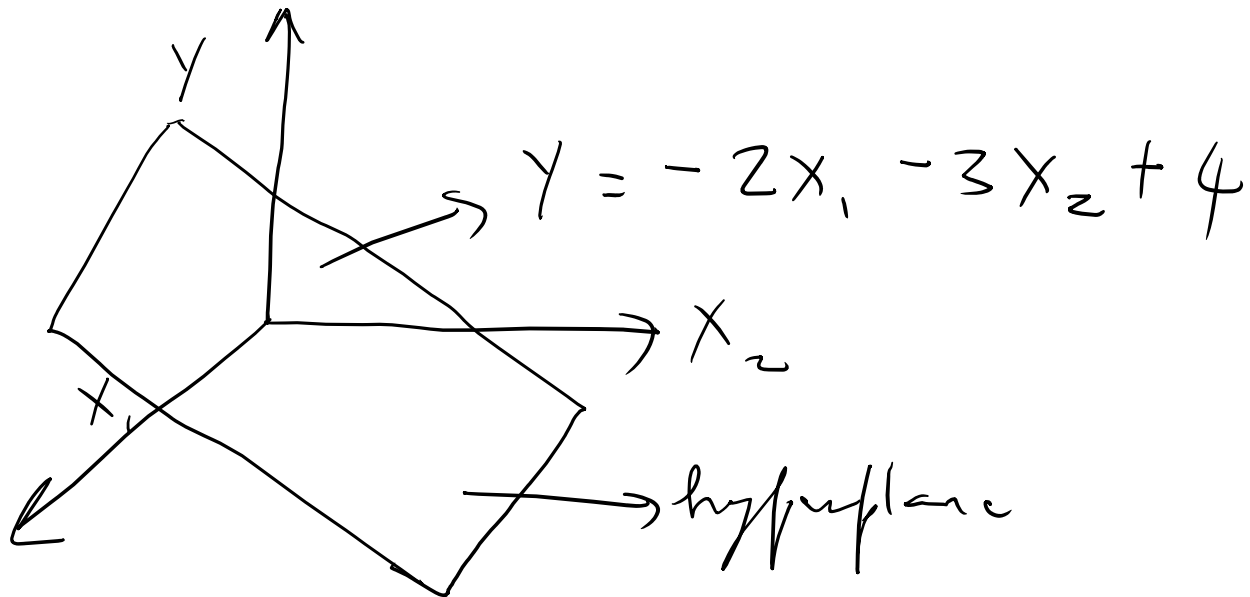
$$Y = a + b_1 X_1 + b_2 X_2 + \dots + b_n X_n$$

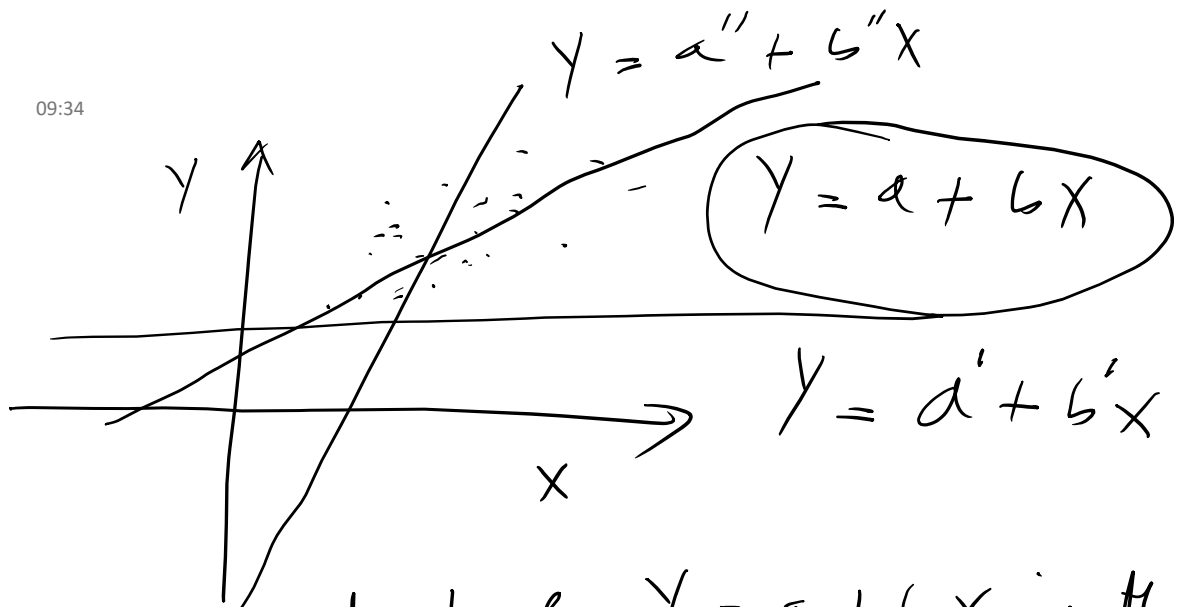
↑

weekly gross revenue

$$\frac{dY}{dX_1} = b_1$$

$$\frac{dY}{dX_2} = b_2$$





Intuitively, $Y = a + bX$ is the best fit

$\sum (Y_i - \hat{Y}_i)^2$ is minimized when
 $\hat{Y} = a + bX$ and not
 when $\hat{Y} = a' + b'X$ or $a'' + b''X$
 how do we find the optimal a & b ?

Minimize $E_{vv} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$

As a & b change y_i will
also change.

$$E_{vv} = E = f(a, b)$$

$\frac{\partial E}{\partial a} = 0$ $\frac{\partial E}{\partial b} = 0$ will give you
two equations in a & b whose
solution is the optimal setting
for a & b .

$$\hat{Y}_i = a + b X_i$$

$$\text{Error} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = E$$

$$\text{Error} = \sum_{i=1}^n (Y_i - \underbrace{a + b X_i}_{\hat{Y}_i})^2 = E(a, b)$$

$$\frac{\partial E}{\partial a} = 0 = \sum_{i=1}^n \frac{\partial (Y_i - a - b X_i)^2}{\partial a} = 0$$

$$= \sum_{i=1}^n 2(Y_i - a - b X_i)(-1) = 0$$

$$\Rightarrow \sum_{i=1}^n (Y_i - a - b X_i) = 0$$

$$\sum_{i=1}^n Y_i = \sum_{i=1}^n a + b \sum_{i=1}^n X_i \quad - (1)$$

$$\underbrace{\sum_{i=1}^n Y_i}_{\text{known}} = \underbrace{na}_{\text{unknown}} + b \underbrace{\sum_{i=1}^n X_i}_{\text{known}} \quad - (1)$$

this is a linear equation in 2 & 6

$$\frac{\partial E}{\partial b} = 0 \text{ gives us}$$

$$\frac{\partial}{\partial b} \sum_{i=1}^n (y_i - a - b x_i)^2 = 0$$

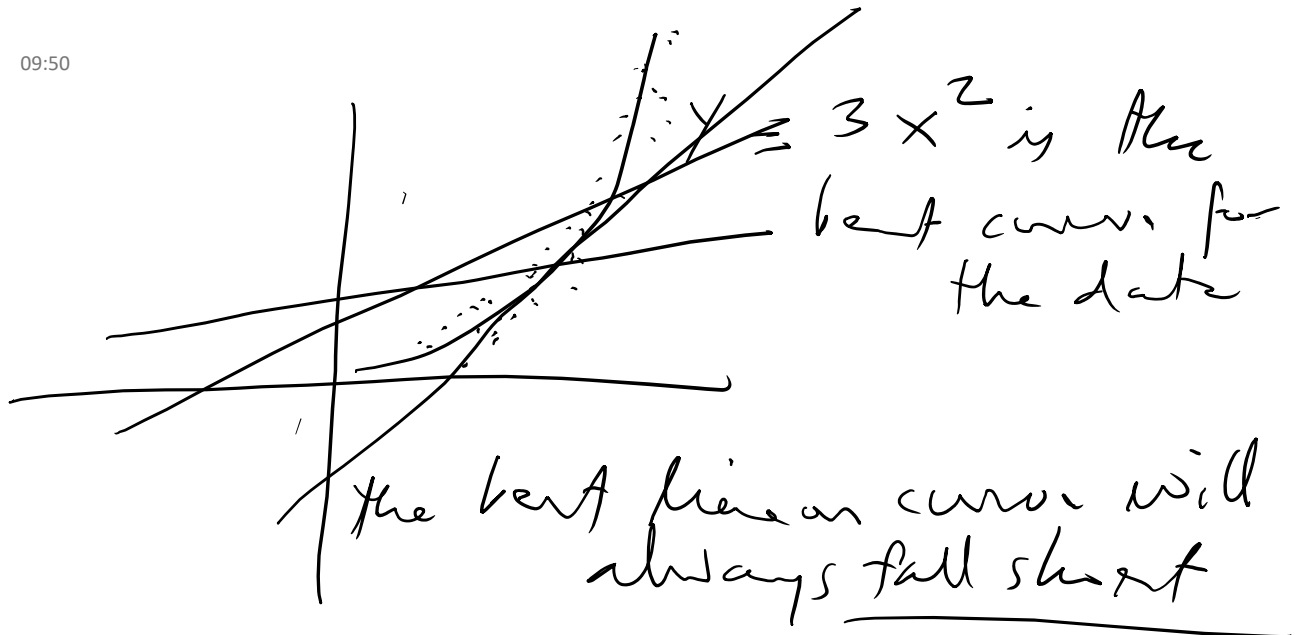
$$\sum_{i=1}^n \frac{\partial}{\partial b} (y_i - a - b x_i)^2 = 0$$

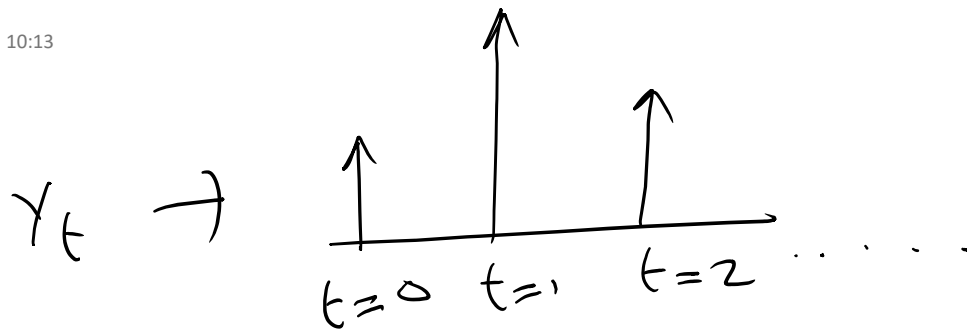
$$\sum_{i=1}^n 2(y_i - a - b x_i)(-x_i) = 0$$

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2$$

constant

→ $\textcircled{2}$
 another linear equation in a
 and b





trend, seasonality, cyclicity,
irregularity and so on

We want to impose a model on data.